Mahdis Sabzi

Architecture portfolio

Selected Academic Work

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Education

B.Sc. in Architectural Engineering
University of Science and Culture, Tehran, Iran

Skills

Revit- Advanced
AutoCAD – Advanced
Photoshop – Intermediate
Lumion, Enscape – Intermediate
3dsmax, Vray - Basic
AI Assisted Visualization & Prompt Design

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In Between
(Residential-
Commercial)
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Children
Hospital
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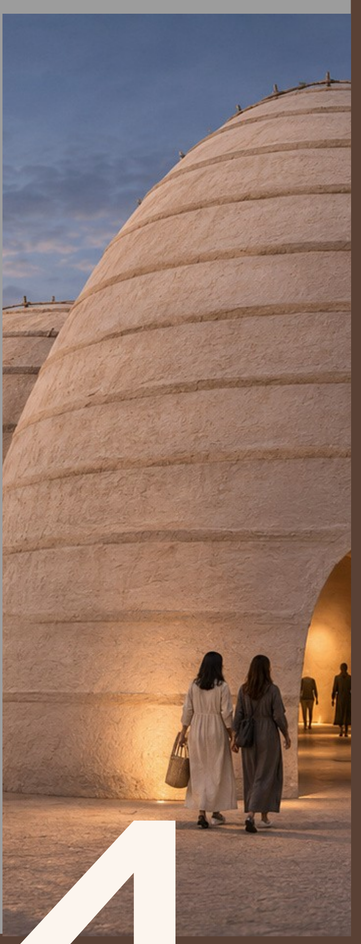
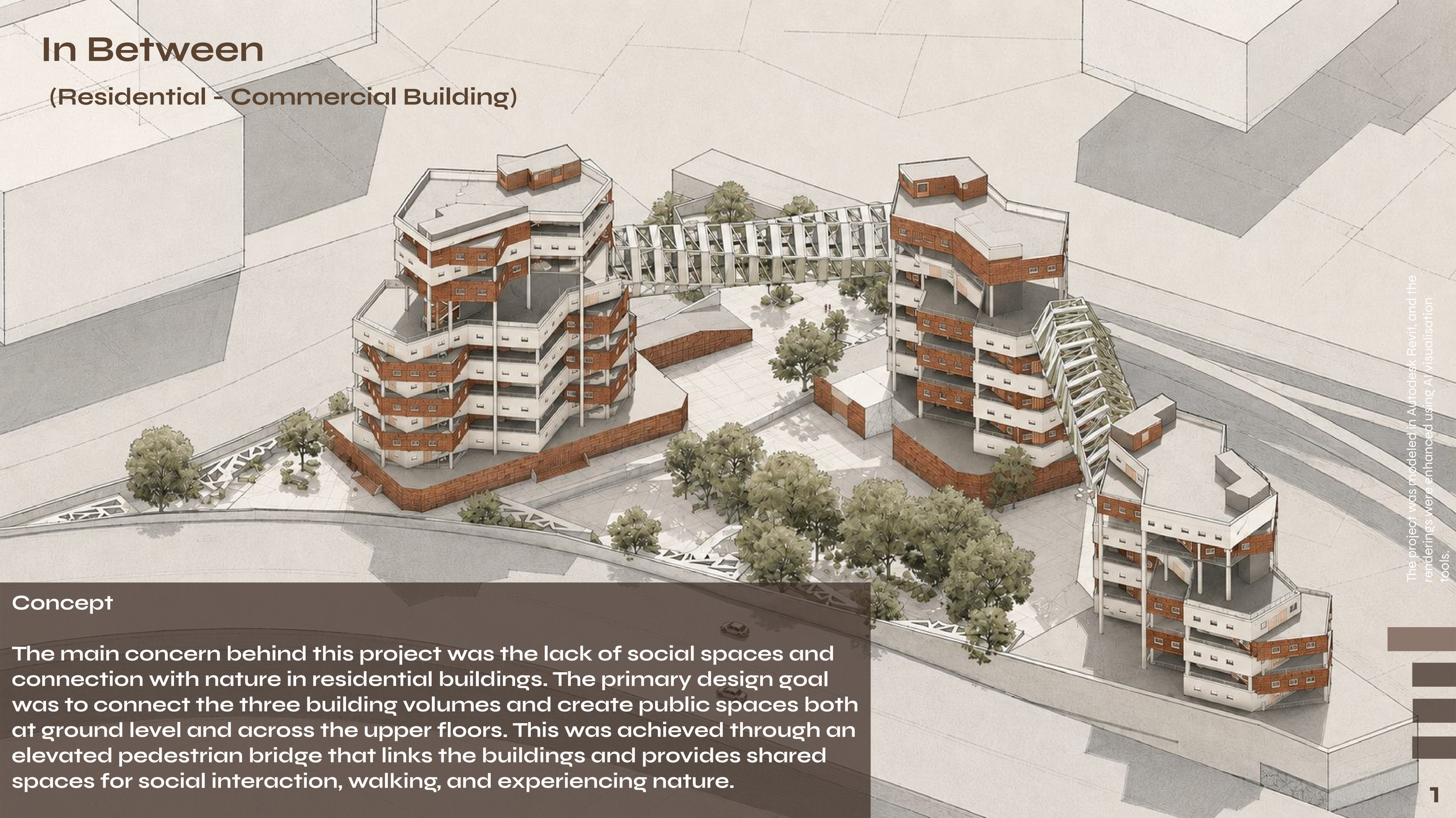


4

Iran Pavilion
32-40

In Between

(Residential - Commercial Building)



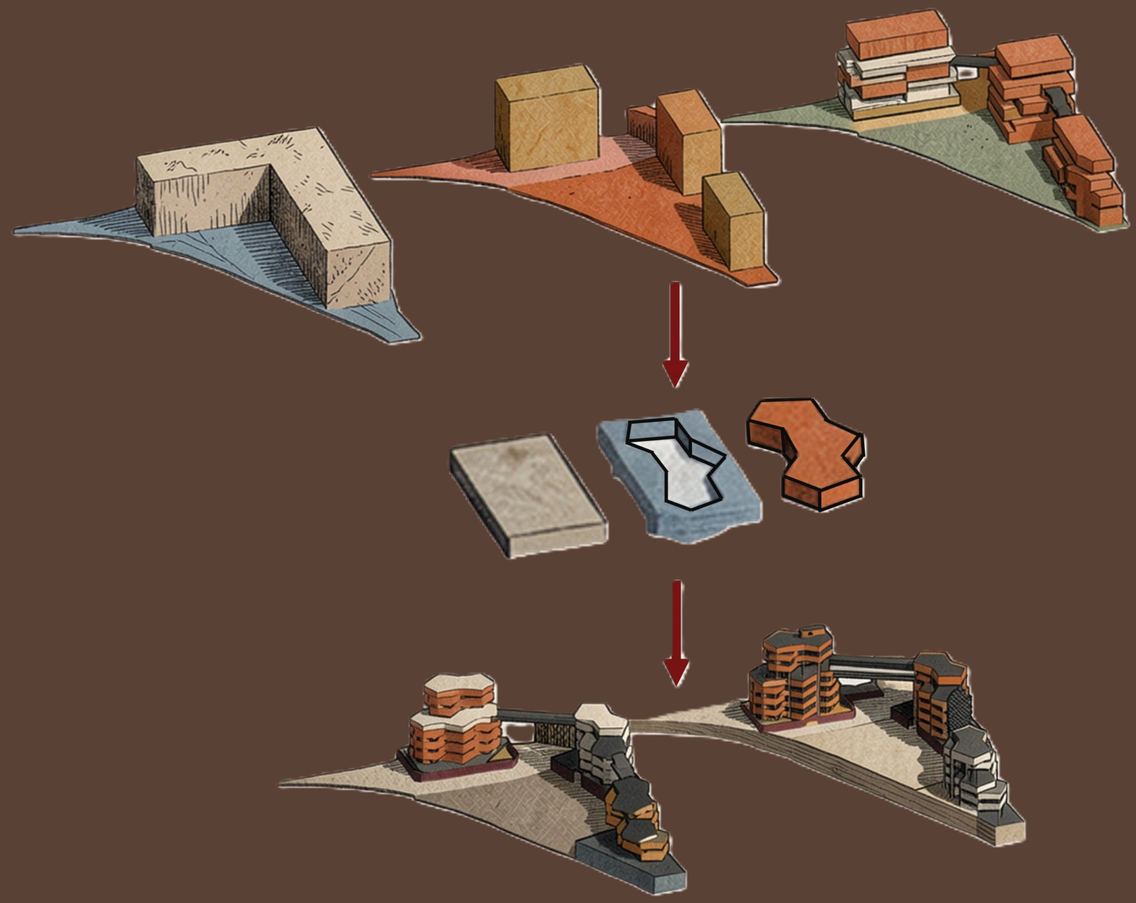
Concept

The main concern behind this project was the lack of social spaces and connection with nature in residential buildings. The primary design goal was to connect the three building volumes and create public spaces both at ground level and across the upper floors. This was achieved through an elevated pedestrian bridge that links the buildings and provides shared spaces for social interaction, walking, and experiencing nature.

The project was modeled in Autodesk Revit, and the renderings were enhanced using AI visualisation tools.

Design Process

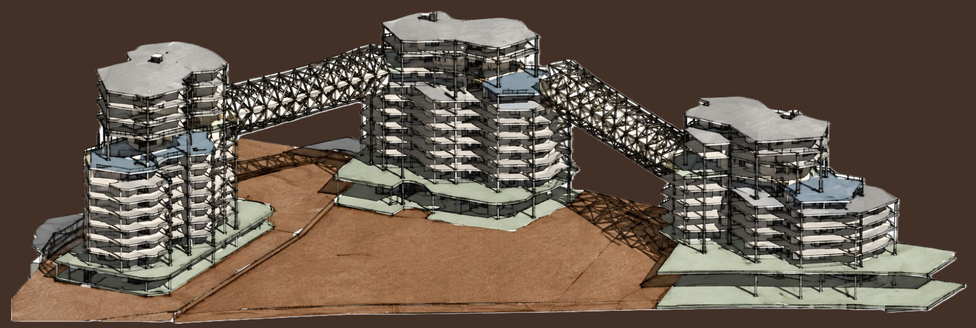
The design process began with shaping three building volumes according to the site geometry. The volumes were arranged to respond to the site’s form and functional requirements. Pedestrian bridges were then introduced to connect the three buildings at their designated public levels, creating a continuous elevated pathway that integrates circulation, social interaction, and shared public spaces throughout the complex.



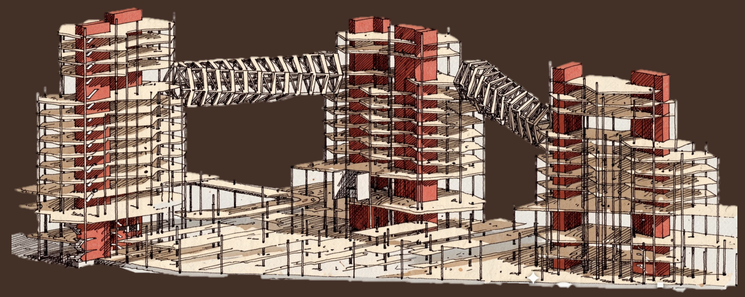
Technical Design

Functional Zoning and Vertical Circulation

White : Commercial
Black: Residential
Yellow: Public Space



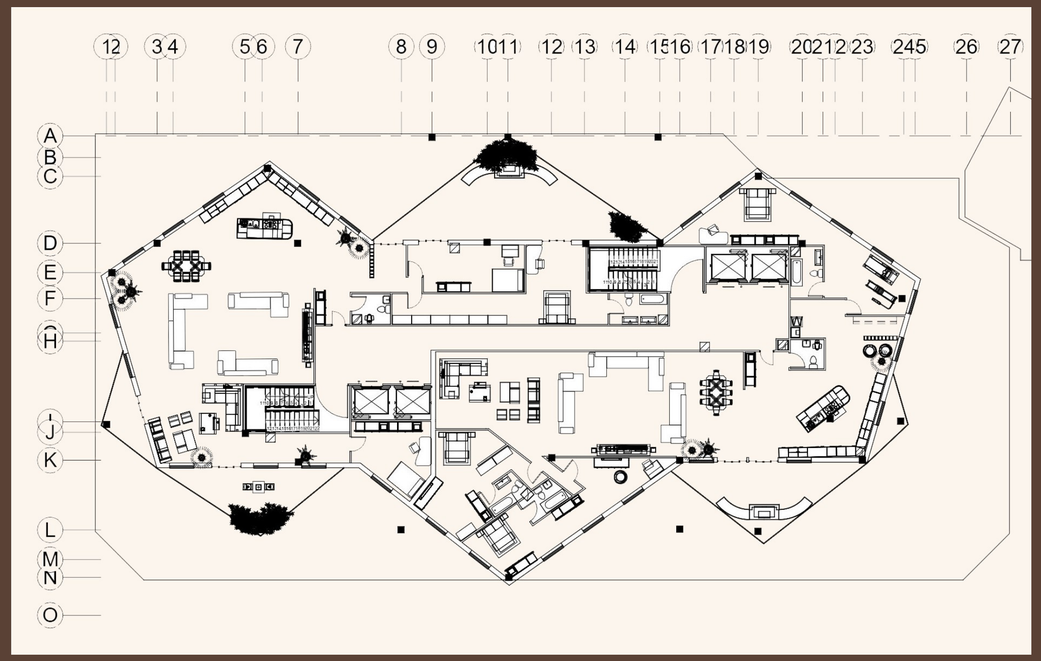
Vertical Access



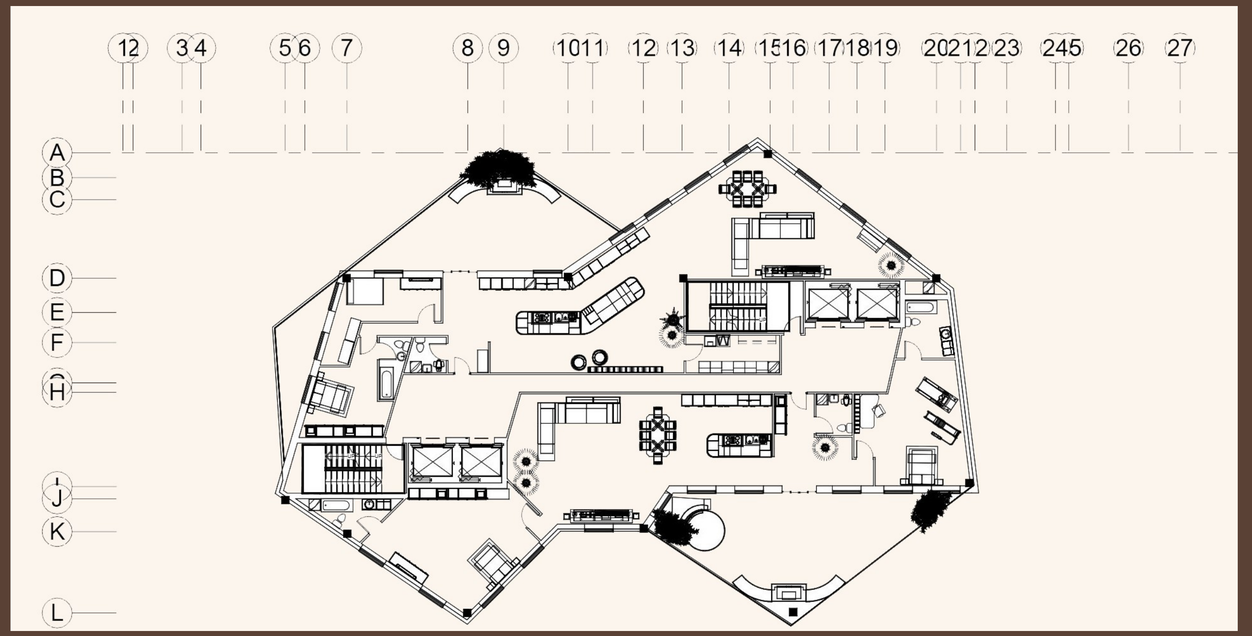
Site Plan

The three building volumes are arranged according to the site geometry. Each residential unit benefits from abundant natural daylight and multiple balconies. The main entrance is located on the northern side of the site to minimize traffic congestion on the main highway. At ground level, landscape design and integrated green spaces create a welcoming public environment for walking, relaxation, and social interaction.

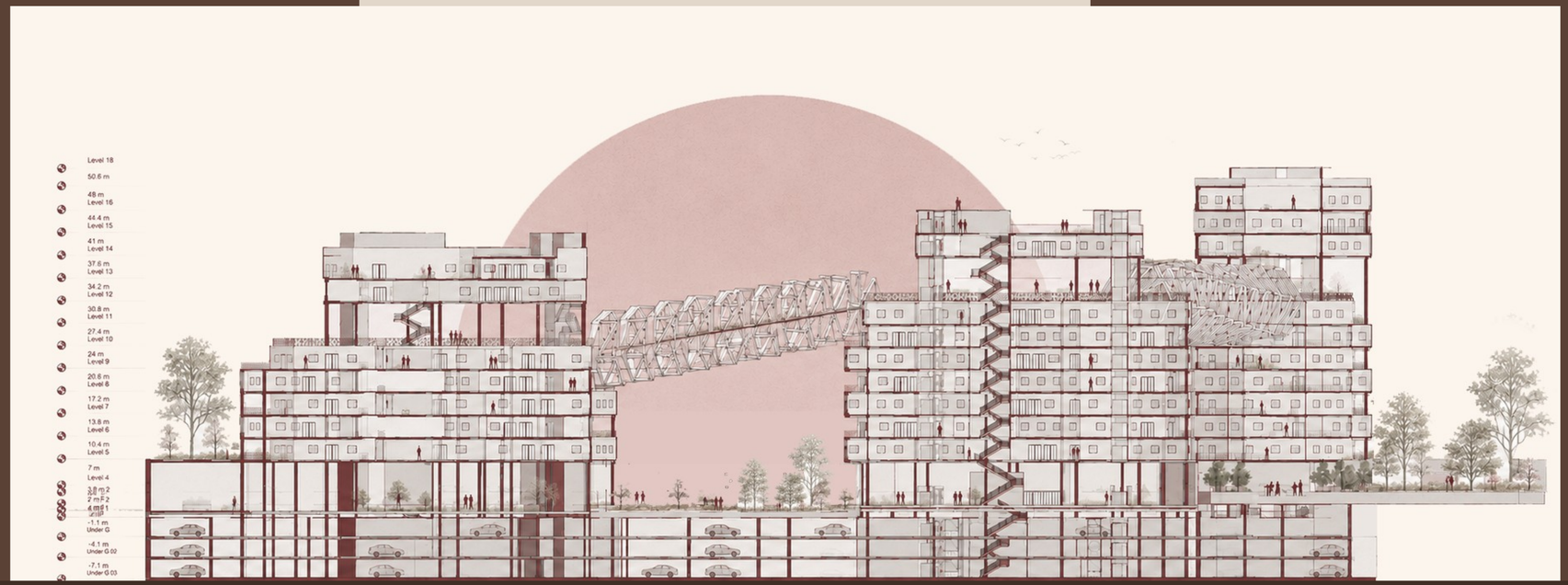
Plans



Sample Plan 1
SC: 1/200

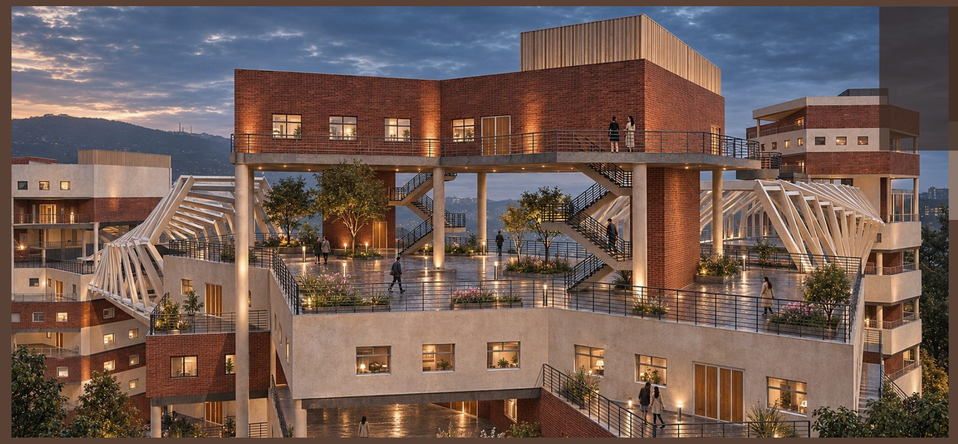


Sample Plan 2
SC: 1/200



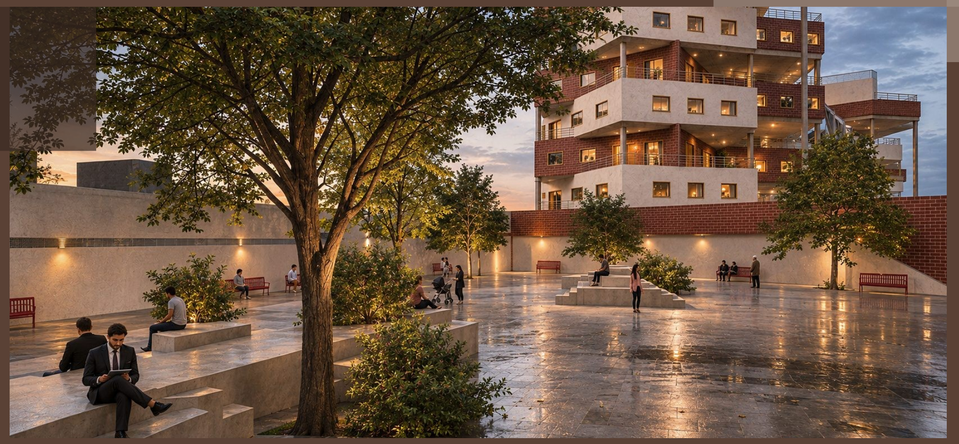
Section A-A
SC: 1/600

Perspective



Public Space

Landscape in Yard



Walking Bridge

Perspective

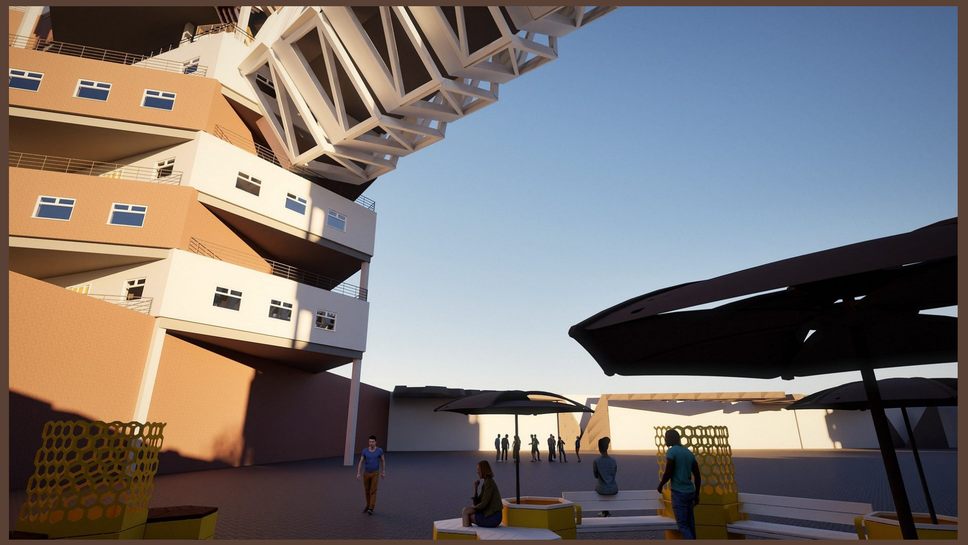
Eye-level Perspective

Pedestrian and
Car Entrance

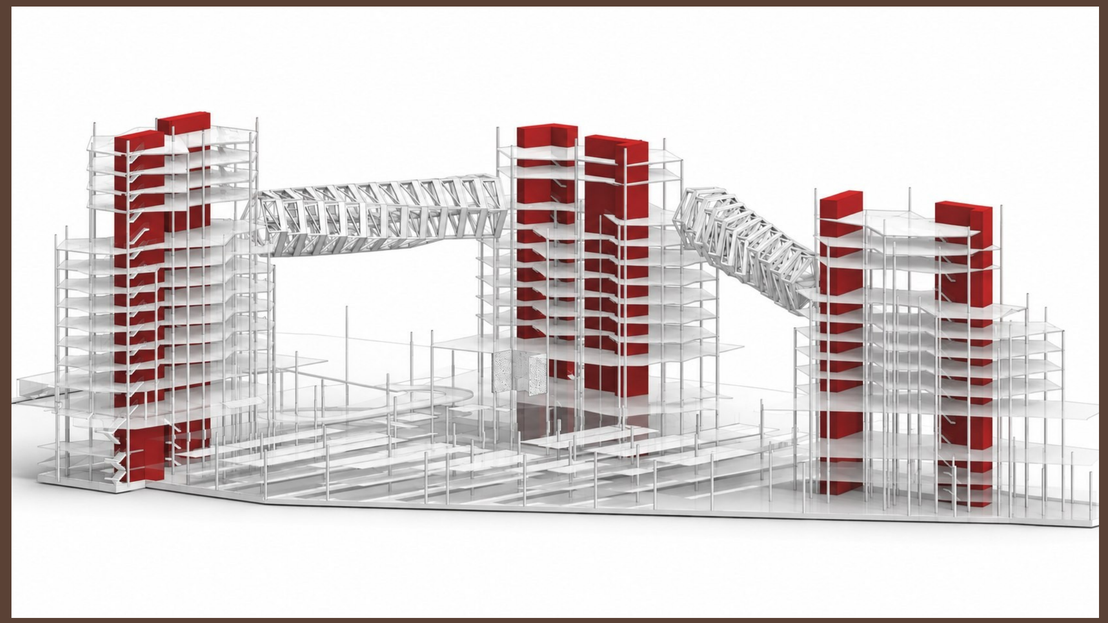
Entrance Ramp and Stairs

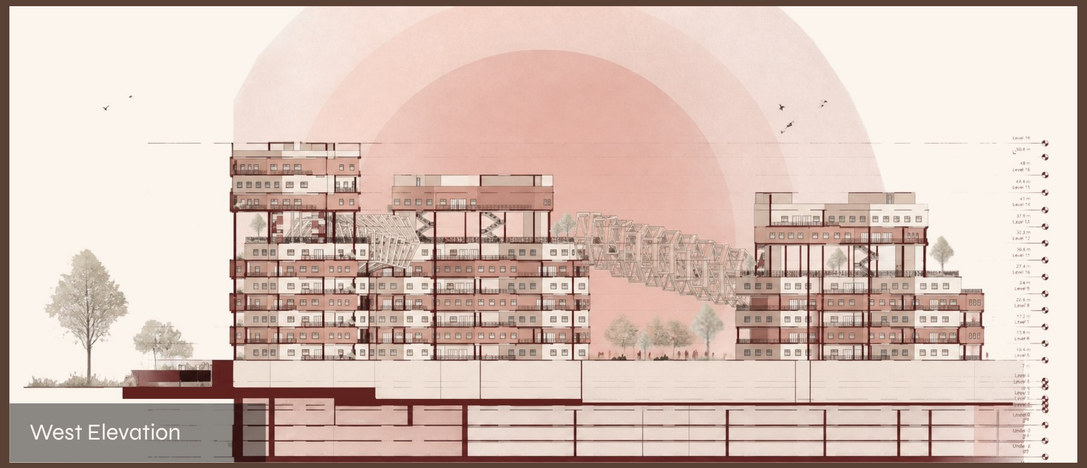
Renders

Rendered from the Revit model in Twinmotion

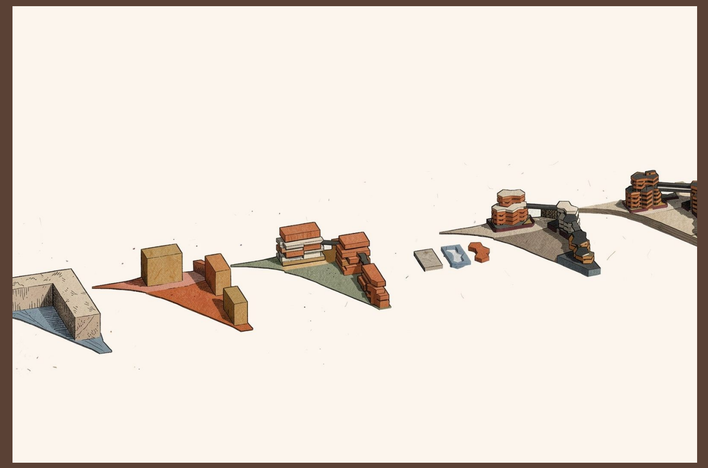


Model





West Elevation



Perspective

AI-assisted visualisations based on the original model

Shared terraces in the last light

The linked volumes from above

Three linked volumes from above

Street corner and connected homes

Perspective

AI-assisted visualisations based on the original model

At the bridge threshold

A lateral view along the bridge

The courtyard from the opposite path

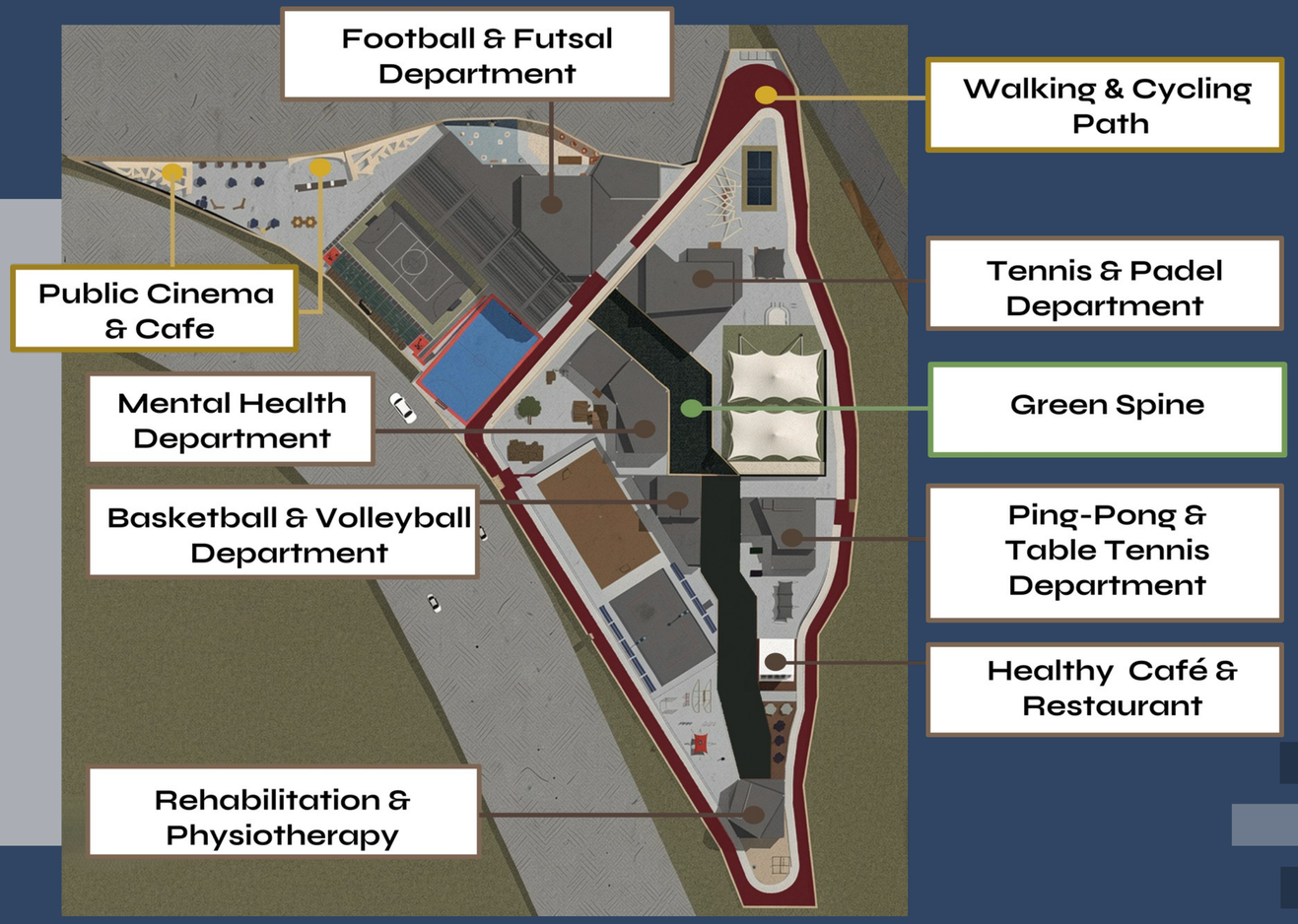
Across the shared terrace

2. Rasta Sport Complex

The Sport and Mental Wellness Center approaches architecture as an essential part of the human well-being experience. Spatial proportions, natural light, landscape, color, and the connection between indoor and outdoor environments are carefully considered to create spaces that support activity, comfort, concentration, and relaxation.

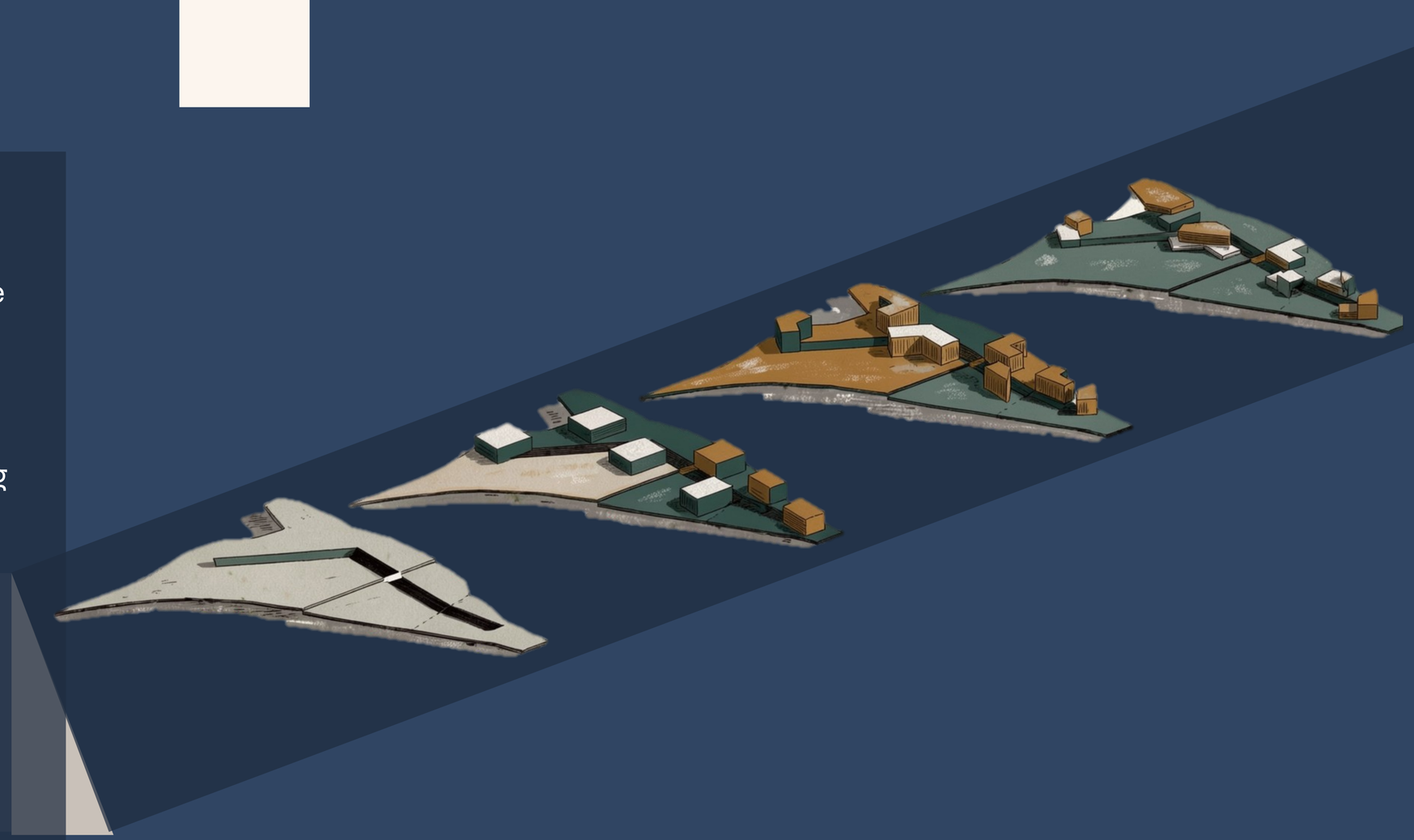
Zoning and Spatial Organization

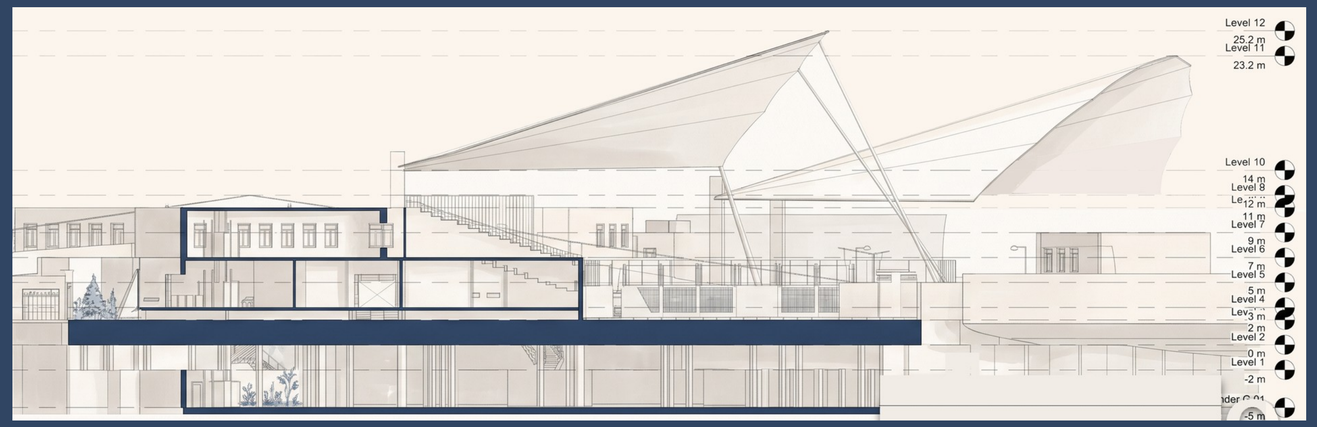
The site is organized into specialized sports and wellness departments, each positioned according to its functional requirements and spatial relationships. The Green Spine acts as the main connecting element, providing a clear circulation route between departments while integrating landscape into the overall design. A peripheral walking and cycling path encourages active movement throughout the site, while outdoor sports facilities and shared amenities enhance accessibility, interaction, and the connection between architecture and nature.



Design Process

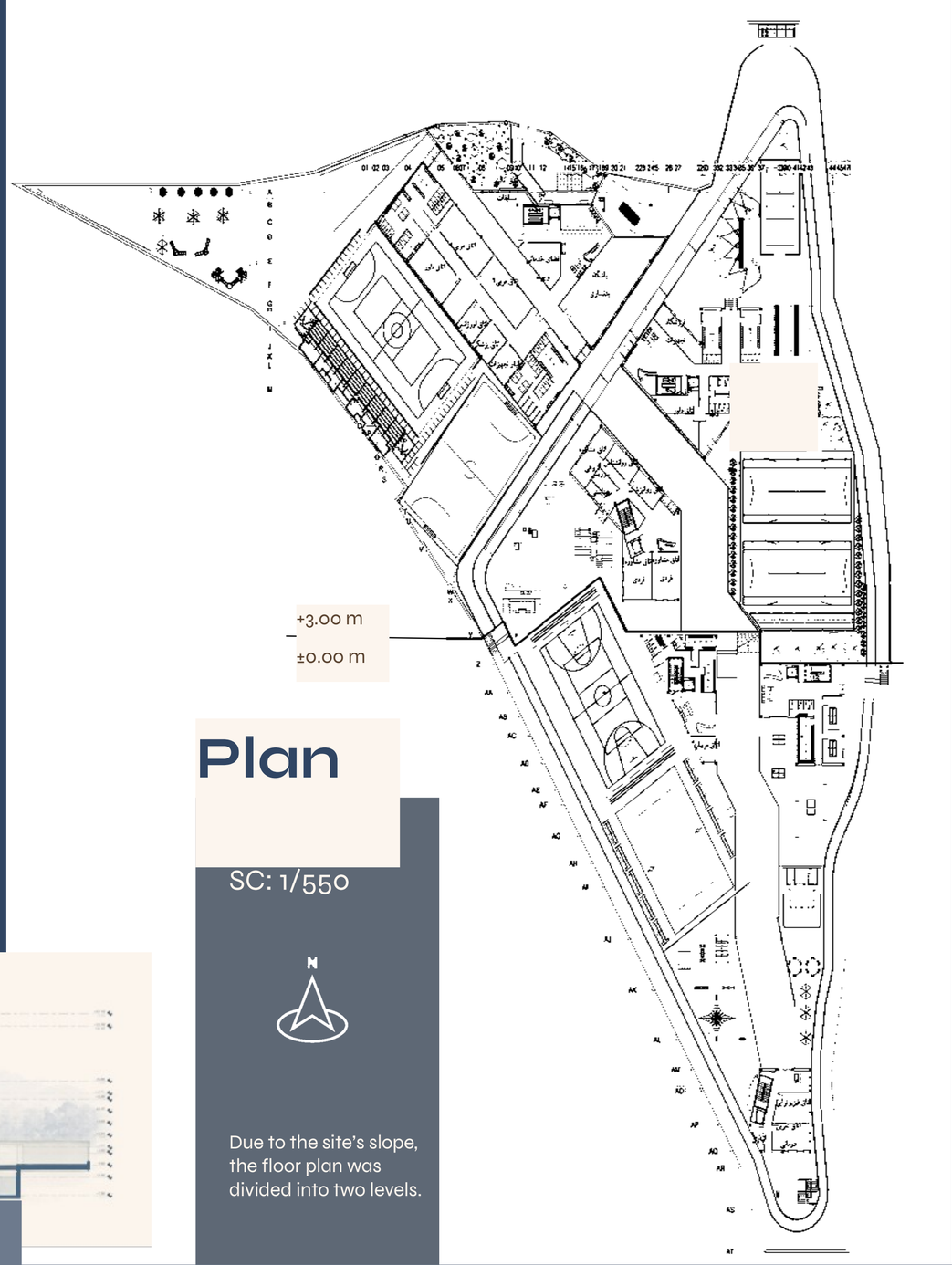
Based on the site shape and the concept of combining architecture with nature, a central Green Spine connects all departments.
The building masses are arranged along this path,
while a loop of walking and cycling paths surrounds
the site. Outdoor sports courts are placed
according to each department's needs,
providing indoor, semi-open, and outdoor spaces with direct access to nature.





Section 1-1 SC: 1/300

Section 2-2 SC: 1/600



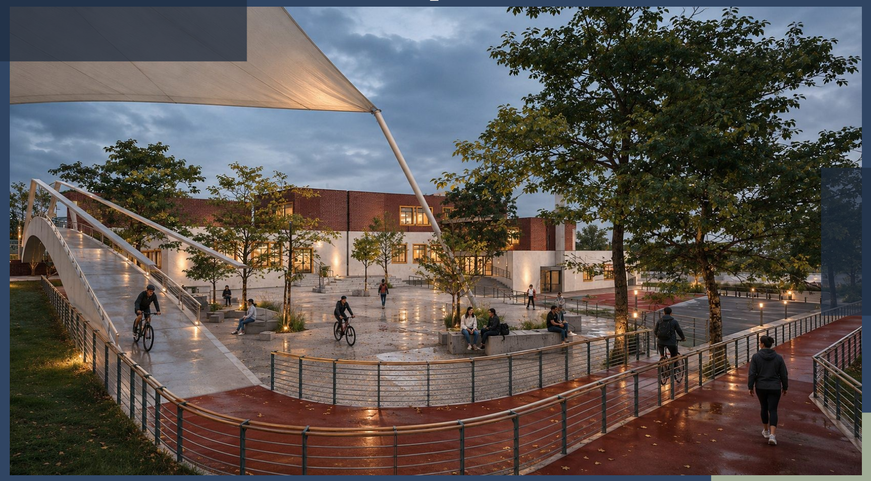
Plan

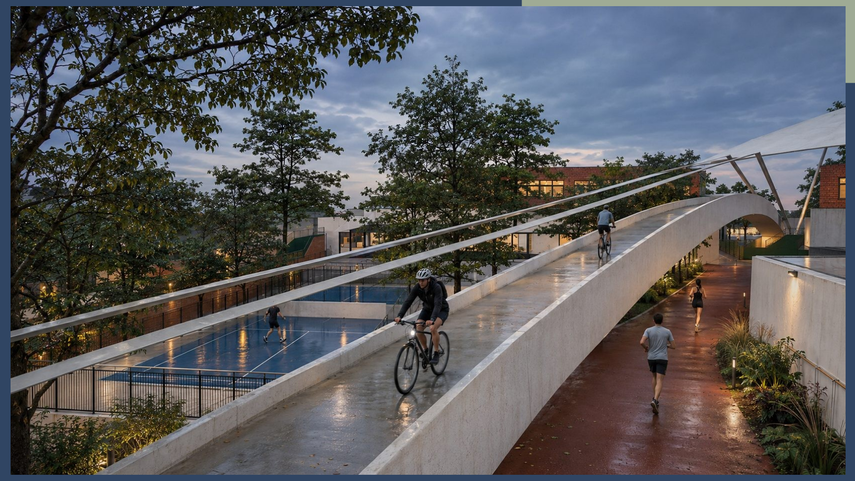
SC: 1/550

N

Due to the site's slope, the floor plan was divided into two levels.

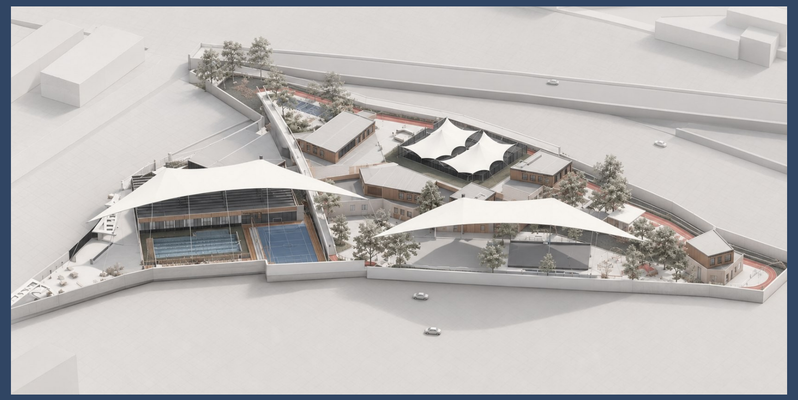
Perspective





Renders

Rendered from the model



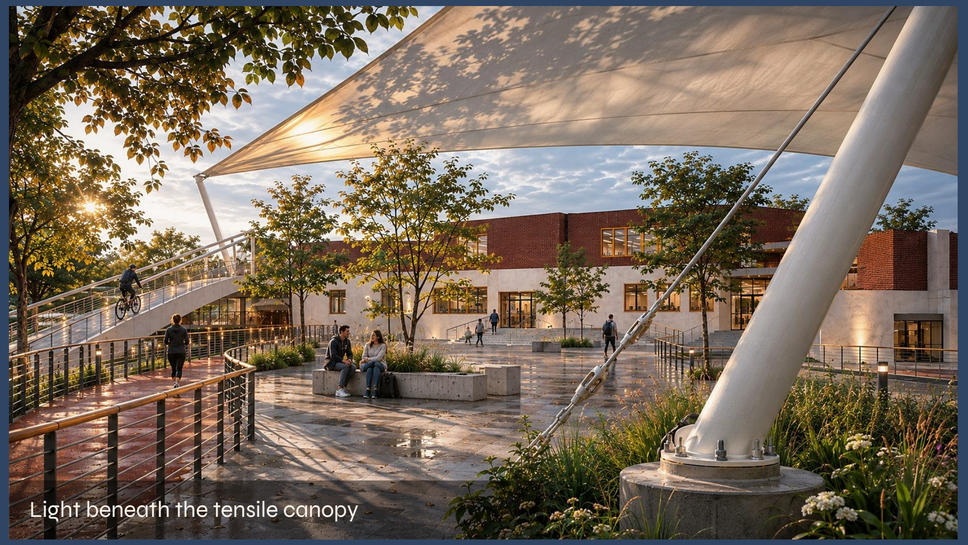
Elevations

East Elevation

West Elevation

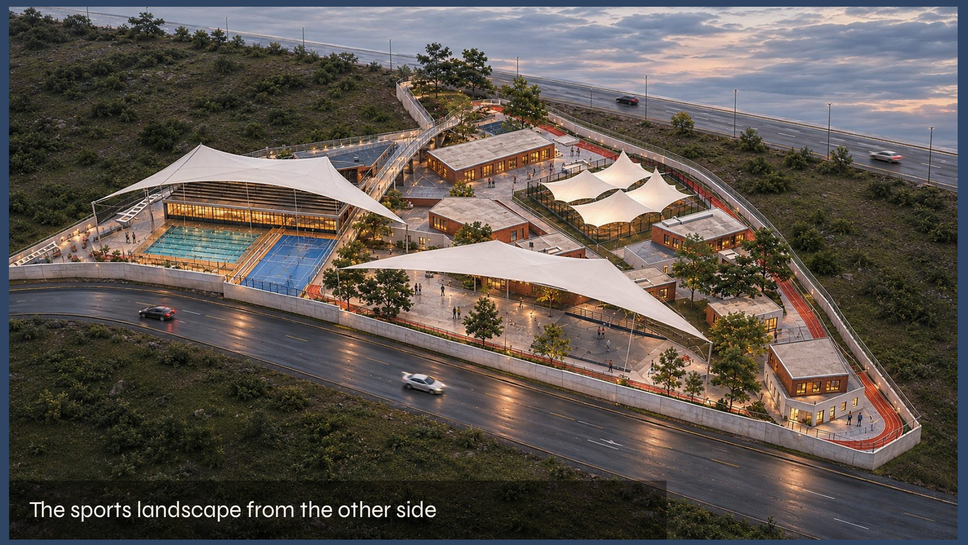
Perspective

AI-assisted visualisations based on the original model



Light beneath the tensile canopy

The Green Spine within the triangular site



The sports landscape from the other side

Arrival along the Green Spine

Perspective

AI-assisted visualisations based on the original model

Basketball and spectator seating

Football under the tensile canopy

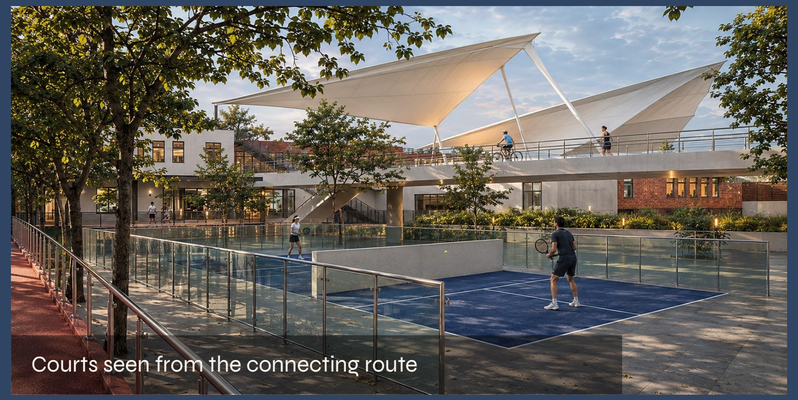
Along the covered football touchline

Bicycle shelter and planted walking route

Perspective

AI-assisted visualisations based on the original model

Under the canopy edge



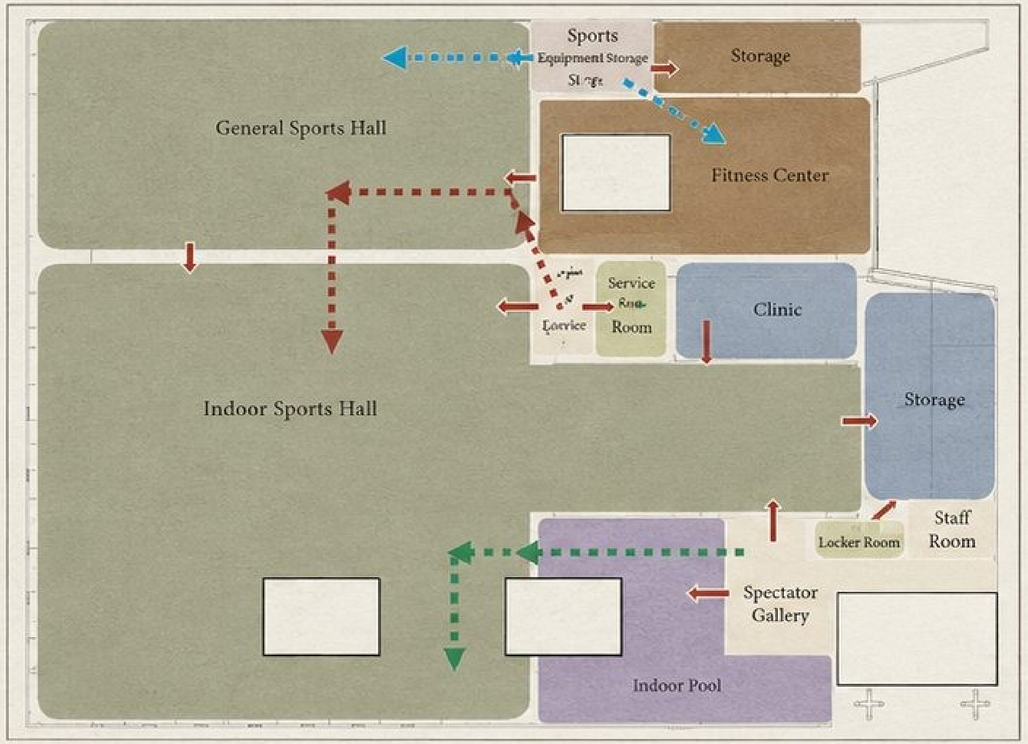
Courts seen from the connecting route

At the bend of the cycle route

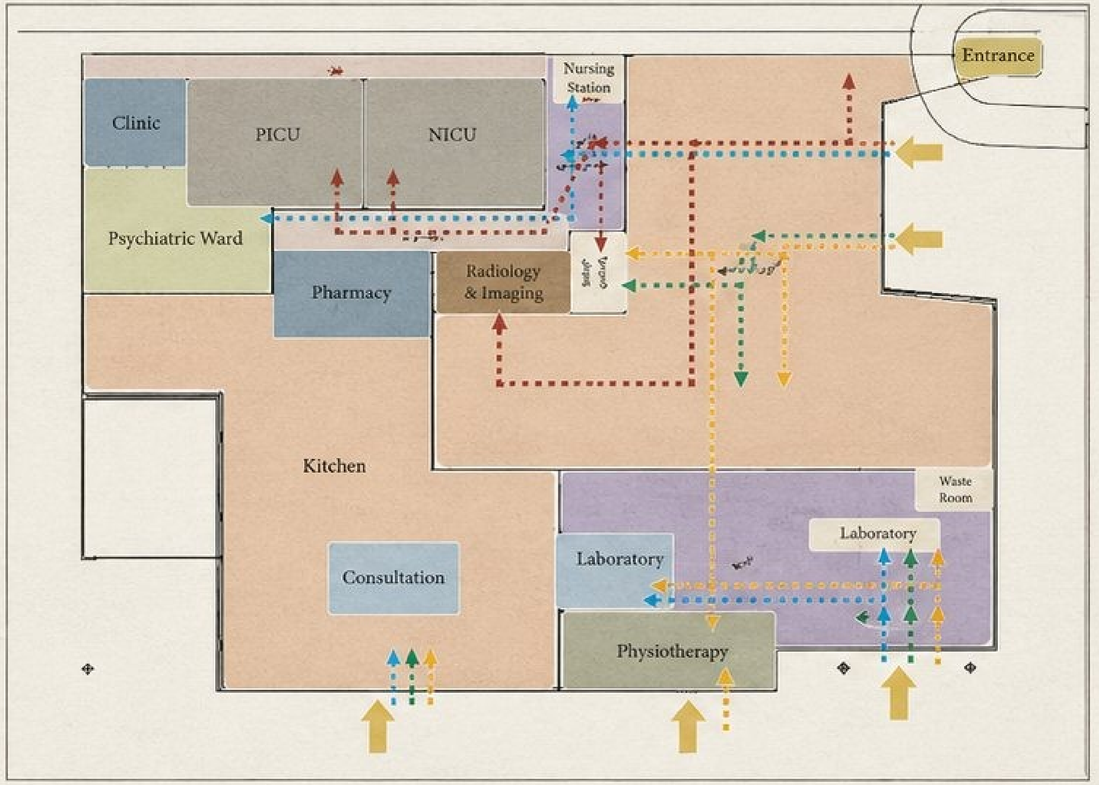
3. Children Hospital

Inspired by LEGO blocks, the project explores modular forms, varied volumes, and vibrant colors to create a playful, child-friendly architectural identity. The concept aims to transform the hospital from a purely clinical environment into a welcoming and engaging space that supports children’s emotional well-being and recovery. Through the integration of natural light, green spaces, play areas, and healing gardens, the design establishes a connection between architecture, nature, and the needs of young patients and their families.

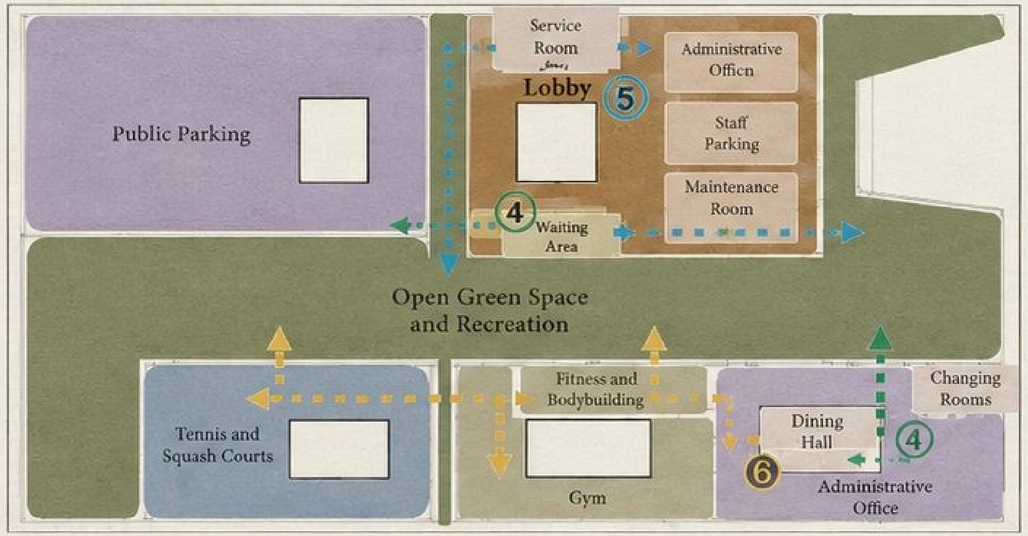
First Floor

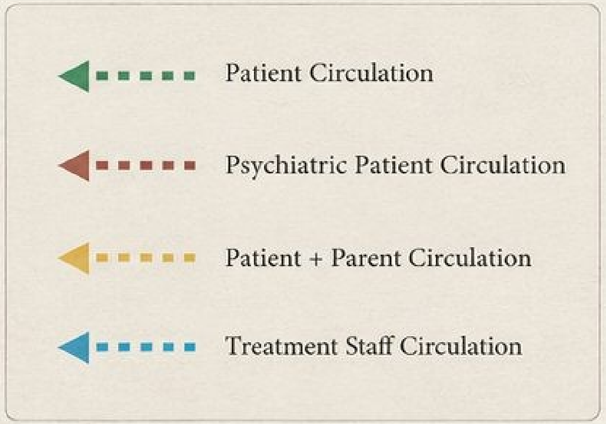


Ground Floor



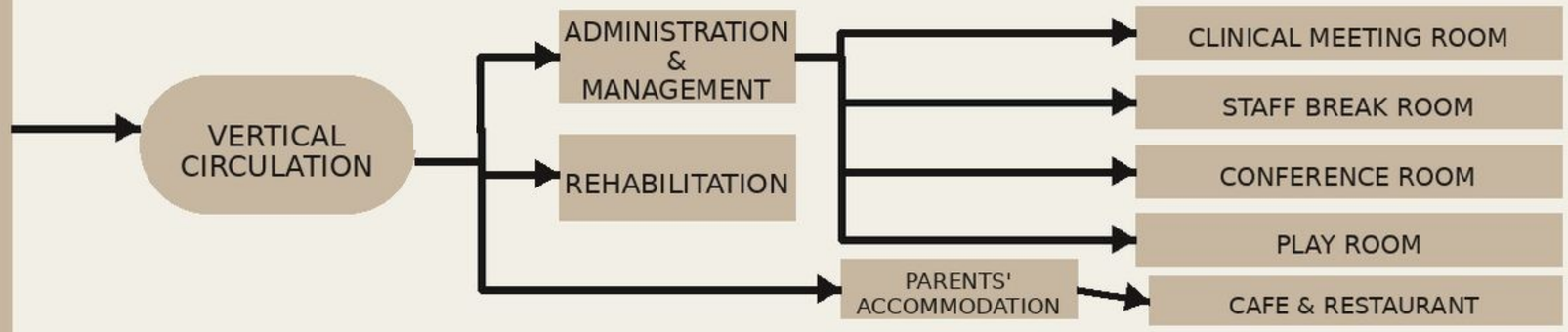
Second Floor



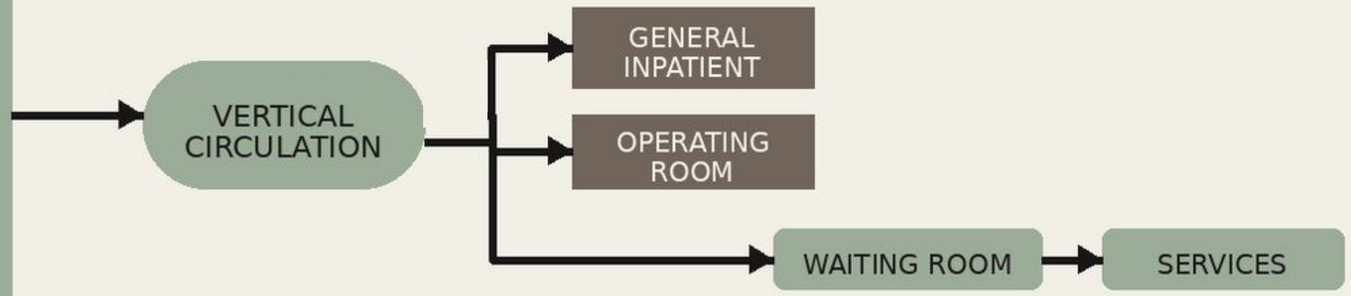


Patient and Staff Circulation Diagram

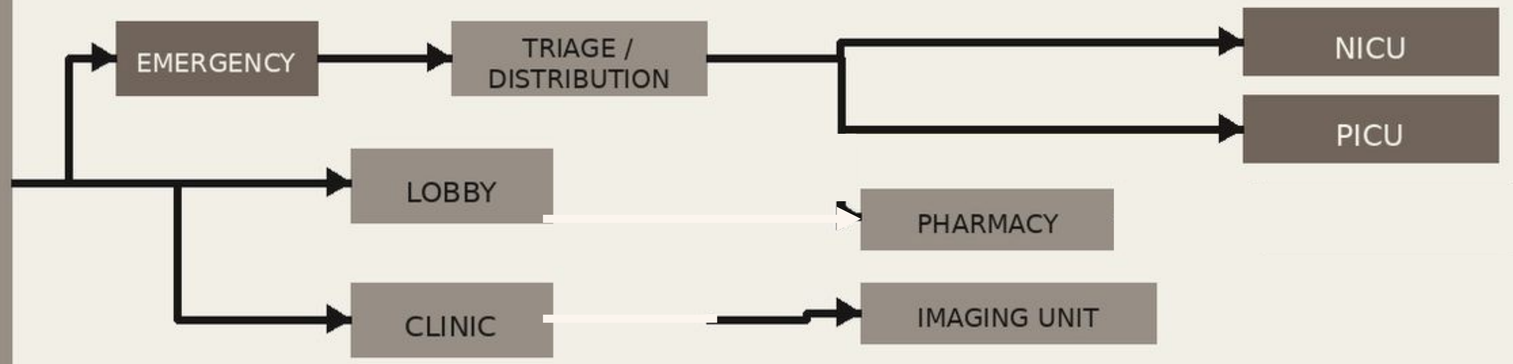
SECOND FLOOR



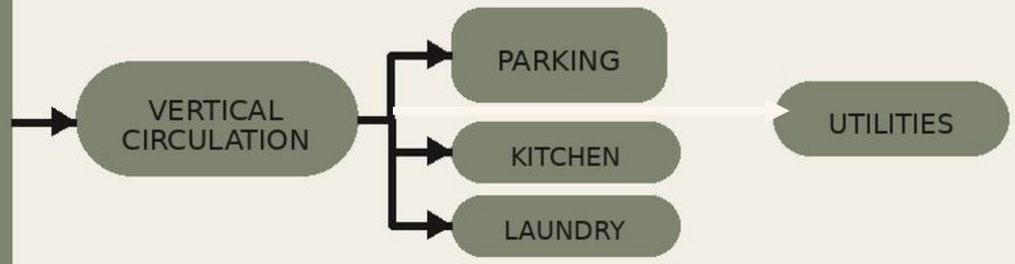
FIRST FLOOR



GROUND FLOOR



BASEMENT



Spatial Relationship Diagram

Perspective

Main Entrance

Emergency Entrance

Building Entrance Elevation

Outdoor Children’s Playground

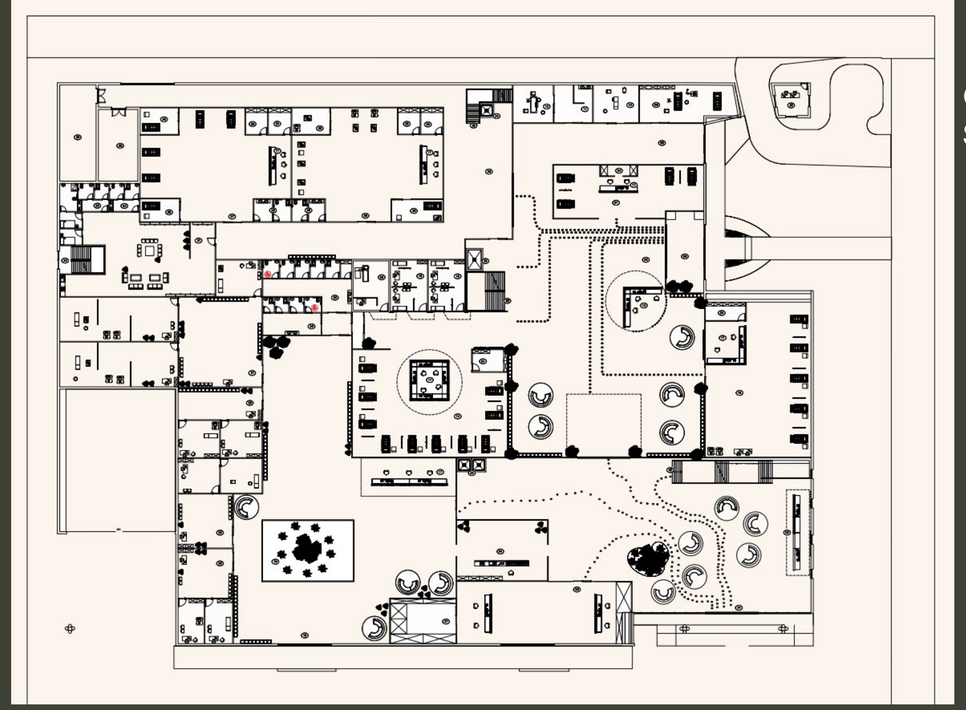
Perspective

Emergency Examination

General Inpatient

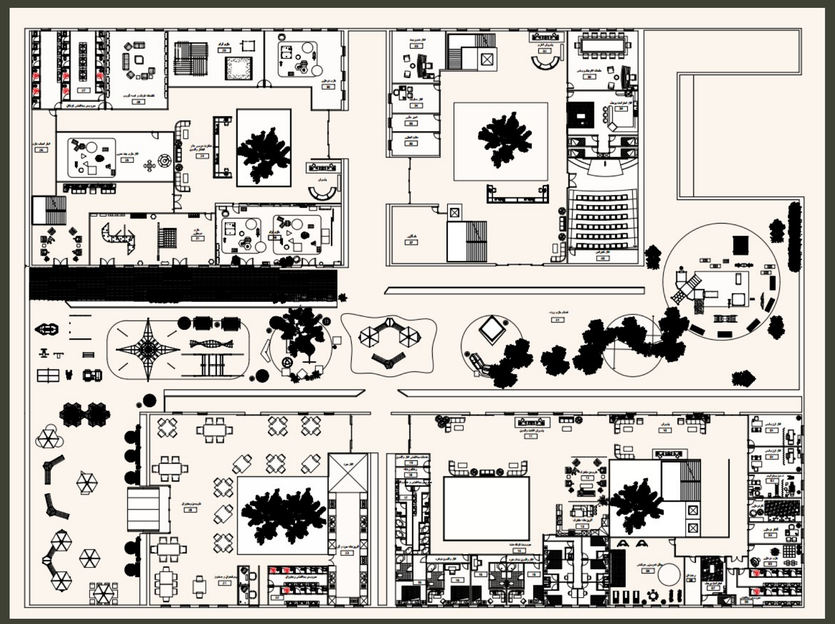
Outdoor Children’s Playground

Plans

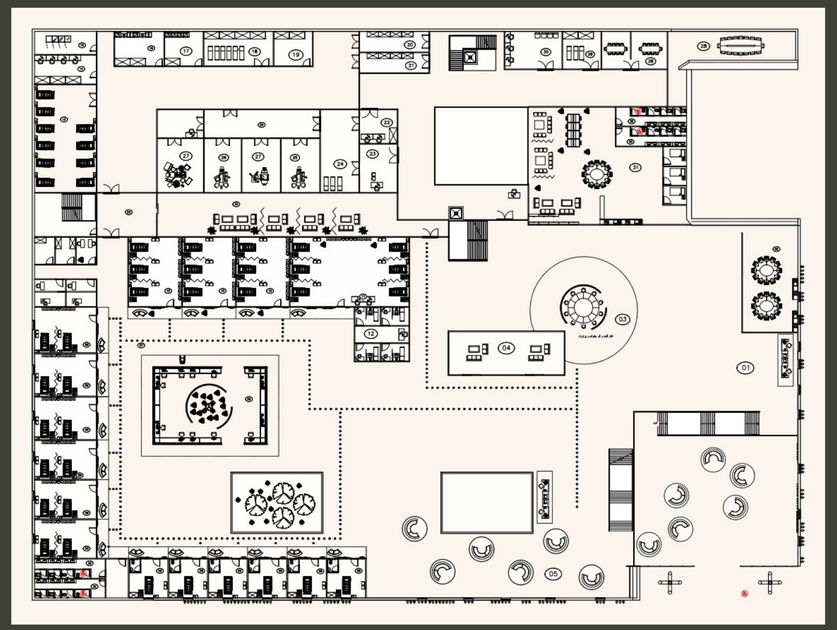


Ground Floor
SC: 1/500

Second Floor
SC: 1/500



First Floor
SC: 1/500



Model

Perspective

AI-assisted visualisations based on the original model

A quiet morning in the healing atrium

Arrival from the garden path

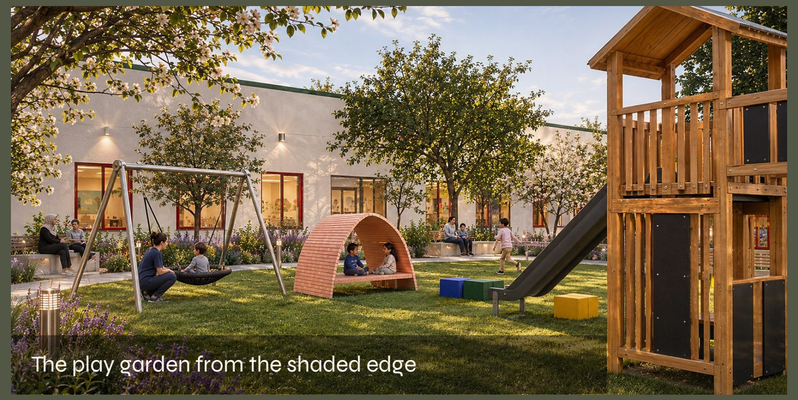
The emergency approach

Through the entrance atrium

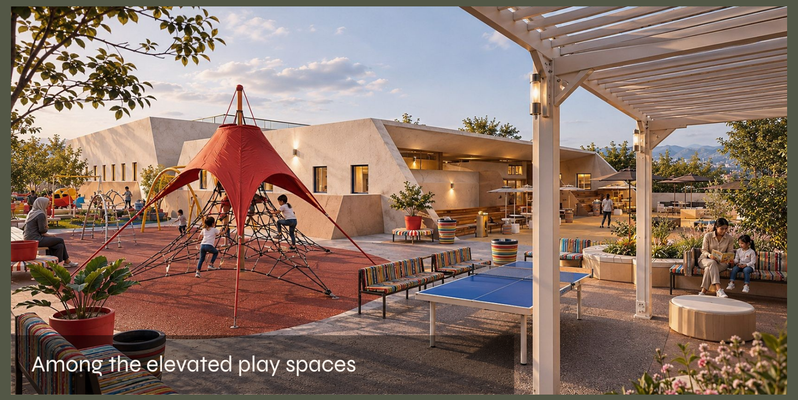
Perspective

AI-assisted visualisations based on the original model

The hall from the upper landing



The play garden from the shaded edge



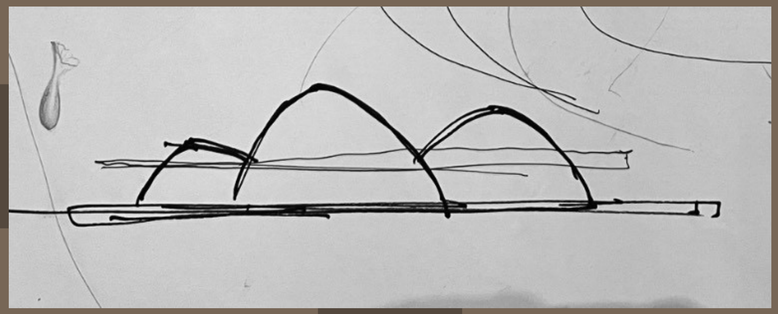
Among the elevated play spaces

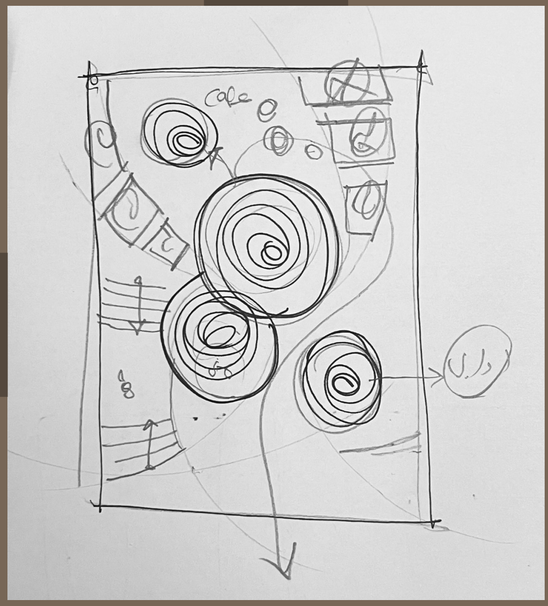
4. Iran Pavilion

The Iran Pavilion at Dubai Expo is inspired by the natural landscape of Kenarak in Sistan and Baluchestan. The project combines traditional Iranian architecture, earth-based construction, and cultural heritage to create an immersive experience. Through interactive galleries, Persian art, poetry, music, pottery, and dance, the pavilion introduces Iran’s rich cultural identity while establishing a connection between architecture, nature, and tradition.

Design Process

The Iran Pavilion is inspired by the unique landscape of Kenarak in Sistan and Baluchestan, where desert dunes, turquoise water, mountains, and vegetation meet. The design combines nature, traditional Iranian architecture, and cultural heritage to create an immersive experience through light, water, sound, and interactive exhibition spaces.





Super adobe Structural System

Super adobe is an earth-based construction technique developed by Iranian architect Nader Khalili. It uses long polypropylene bags filled with compacted earth to create load-bearing walls and dome-shaped structures. This system connects contemporary architecture with traditional Iranian building methods while offering thermal mass and the potential to reduce reliance on conventional construction materials.

Entrance

The entrance features a domed ceiling decorated with traditional Iranian mirror work. Reflective surfaces and a water feature create dynamic light effects, welcoming visitors into the pavilion and introducing its cultural identity.

Galleries

Shahr-e Farang Gallery

Inspired by a traditional Iranian visual entertainment device, this gallery uses small wall openings to reveal scenes of Iranian culture and history. Subtle lighting, an underground water feature, and a candle-lighting niche create a mysterious and interactive experience.

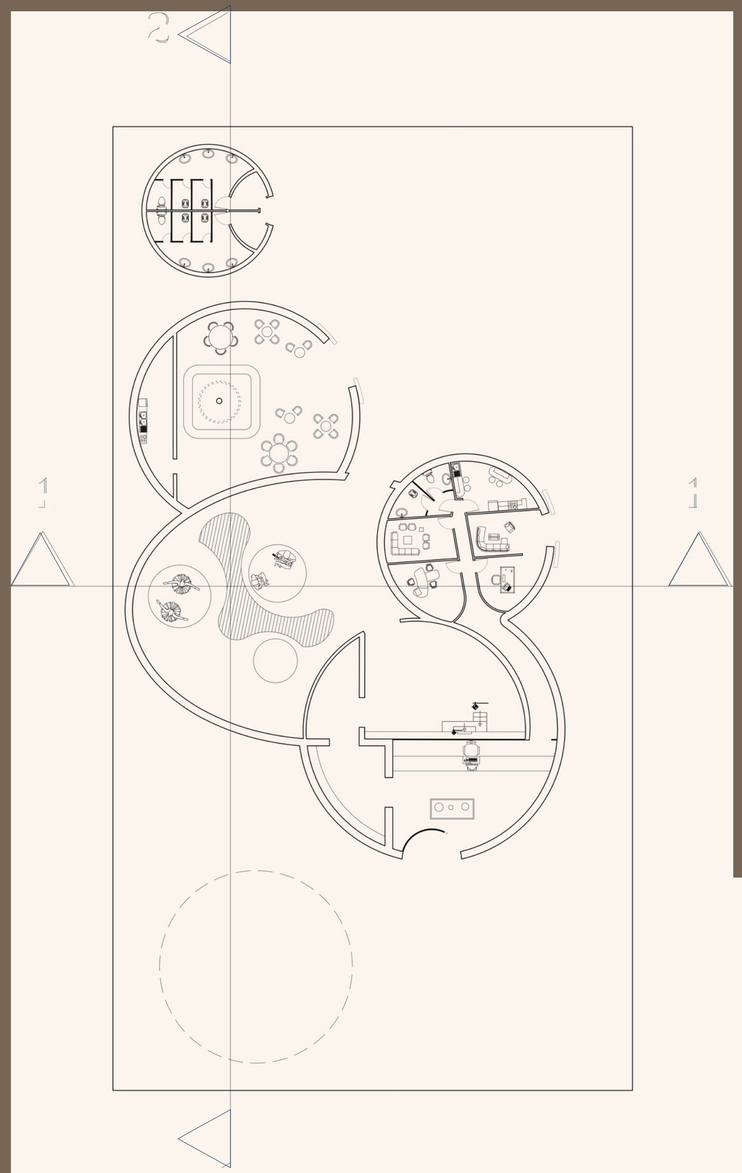
Galleries

Carpet Gallery

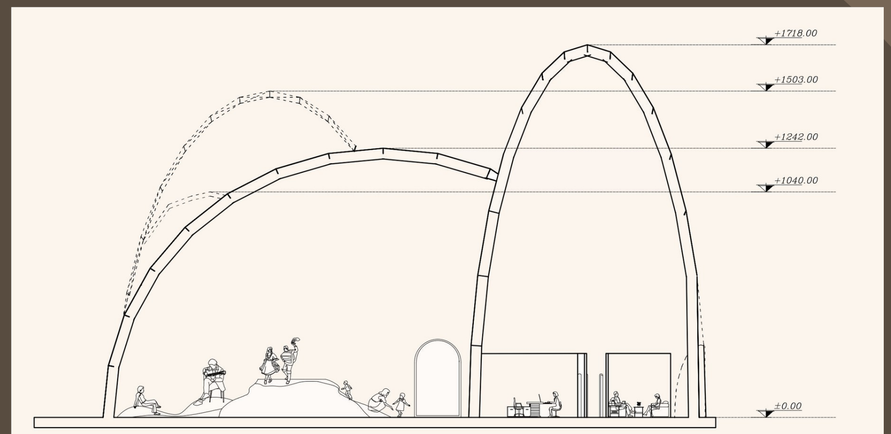
This L-shaped corridor is inspired by traditional Iranian houses. A Persian carpet is integrated into the wall for visitors to touch, while natural light, candlelight, and floral fragrances create a multisensory introduction to Iranian craftsmanship.

Kenarak Gallery

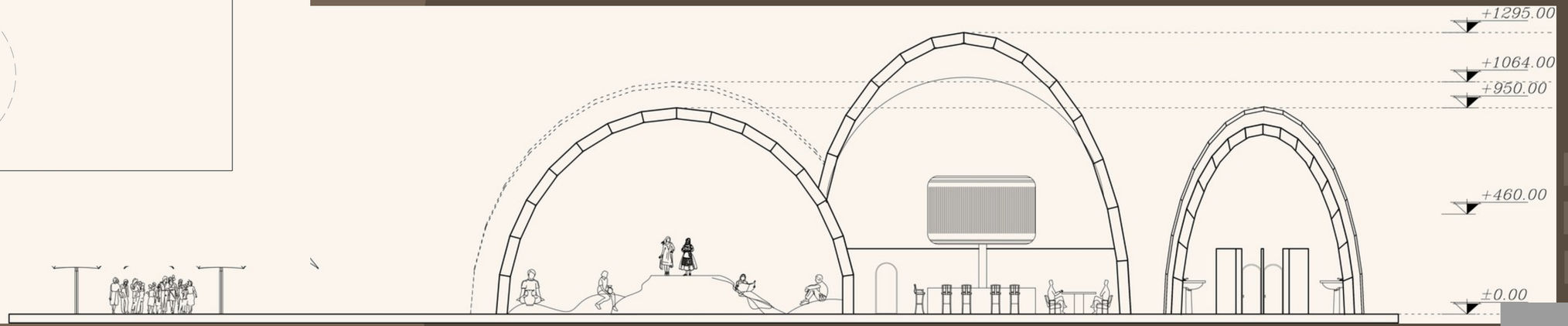
The Konarak Gallery features four earthen hills, each representing an aspect of Iranian culture: Persian poetry, traditional pottery, Iranian music, and folk dance. Visitors walk barefoot through water and sand, interact with performers, listen to poetry and music, and experience traditional crafts. The hills are accessible, allowing visitors to climb up and sit alongside performers, creating an immersive cultural experience inspired by the coastal atmosphere of Kenarak.



Plan
SC: 1/100



Section 1-1
SC: 1/170



Section 2-2
SC: 1/130

Perspective

AI-assisted visualisations based on the original model

Mirrorwork and water at the entrance threshold

A low oblique view across the shells

The earthen shells from the opposite approach

Perspective

AI-assisted visualisations based on the original model

Earthen shell profiles at dusk

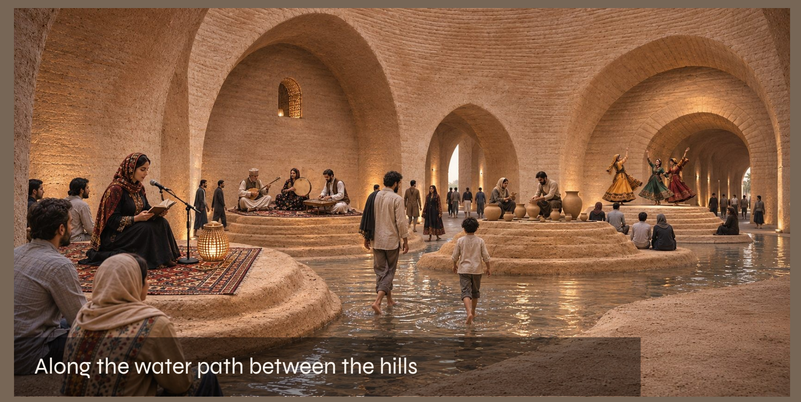
Mirror reflections at the entrance water edge

Viewing niches along Shahr-e Farang

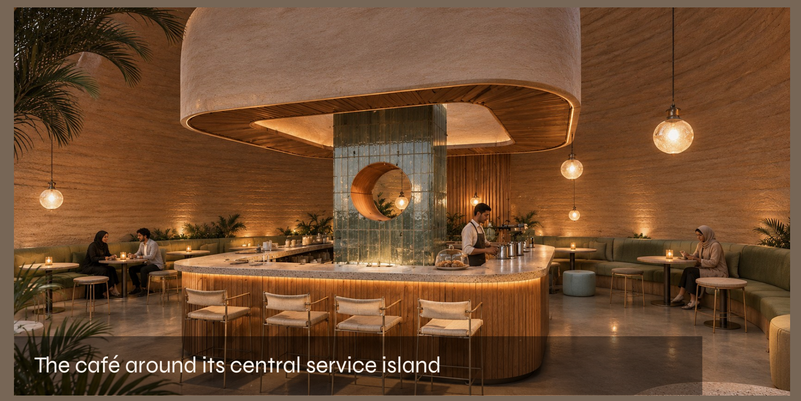
Perspective

AI-assisted visualisations based on the original model

The wall carpet around the gallery turn



Along the water path between the hills



The café around its central service island

The End
Thanks for your time.

Mahdis Sabzi

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